L3.5 Linear independence - Part 2

Transcript Language

English

Hello, and welcome to the online B.Sc. program on data science and programming.

In this video we are going to talk about linear independence.

So, this continues from our previous video on the same topic.

So, let us recall that linear independence means a set of vectors being linearly independent

means that they are not linearly dependent, which is to say that if you take a linear

combination which equals 0, then the only way that can happen is if the coefficients are 0.

So, just to recall, here is the last example that we did in our previous video.

So, we have these three vectors, 1, 1, 2, 1, 2, 0 and 0, 2, 1 in R3 and we take unknown

coefficients a, b and c for these vectors.

So, a times 1, 1, 2 plus b times 1, 2, 0 plus c times 0, 2, 1 is 0, 0, 0.

This is what we assume.

And then we try to work out a, b and c.

And indeed, if we equate the coefficients and from this equation here and write down

the equations in a, b and c and solve them, we get a is equal to b is equal to c is 0.

So, we can now ask the same question in general, how do we check if you have a set of n vectors

in Rm when are they linearly independent?

So, just to, let us go back one step and ask what we did in R3.

So, in R3 what we did is we took arbitrary coefficients.

I will underline the word arbitrary or unknown.

And then checked when, what are the possible solutions for these coefficients for this equation to be 0.

This is a general template.

So, in Rm you will do the same thing.

So, suppose you have coordinates v1j, v2j, vmj, it is in Rm, so there are m coordinates,

so you take the jth vector, that is your, that is vj and you write down the coordinates,

so that is v1j, v2j, vmn and you will do this for each of your vectors v1, v2, vn.

So, let us write the linear combination of these vectors with arbitrary or unknown coefficients

a1, a2, an.

So, here we want to determine the coefficients, which yields this equation a1v1 plus a2v2

plus anvn equals 0.

So, we equate this linear combination on the left to 0.

So, now, we can, this is in Rm remember, so both sides we can express in terms of coordinates.

So, the first coordinate on the left is a1v11 plus a2v12 plus anv1n.

The first coordinate on the right hand side is 0.

That is a 0 vector.

Similarly, for the second coordinate a1v21 plus a2v22 plus anv2n on the left, on the

right it is 0, and we can do this all the way up till m.

So, remember, there are m coordinates.

So, now, you can notice that we have written this system in a different way.

We have written the vij's on the left and the aj's on the right.

Why do we do that?

So, the reason is that here the unknowns are the coefficients a1, a2, an and we, our notation

is that if you have a system ax equals b, so you write summation aij xj.

So, here xjs are ais and aijs are vijs.

That is why we have written it in a slightly altered way.

And now, we know this is a system of linear equations.

In fact, this is a homogeneous system of linear equations, because the right hand side is

0.

And we know exactly how to solve this.

This is something we have done in the previous weeks.

So, the most general method is what we call Gaussian elimination.

If you are in, there are other situations where you can do better by just looking at

determinant and so on.

So, since the ais are arbitrary or unknown, we can treat this like a homogeneous system

of linear equations with coefficients vij and unknowns ai.

This is exactly what we discussed.

So, for linear independence, we have to check if the only choice of ai satisfying the above

identities or equations is ai equals 0 for all i.

So, ai is 0 for each i.

So, remember, we are not, mean when we study linear independence, we are not really interested

in asking what coefficients give you the linear combination 0.

We do not want to expressly evaluate which ai is, what is the set of solutions.

All we want to know is whether the solution is only a set of 0s or whether non-zero solutions

are, do exist, meaning at least one non-zero coefficient, whether such solutions exist.

So, we need not solve the entire system.

That is the point.

So, equivalently in terms of the homogeneous system of linear equations, we have to check

if the only solution is the 0 solution, meaning each of the ais is 0.

So, the conclusion here is, this is the main point, to check if v1, v2, vn in Rm are linearly

independent, we have to check that the homogeneous system of linear equations vx is 0 has only

the trivial solution, meaning the solution where x is 0, that is what we have to check.

And what is this v, v is the matrix we can form out of these vectors, where the jth column

is the vector vj, meaning you take the coordinates of the vector vj and write that in as a column

of v. That is how you get a matrix v.

So, let us do this.

Let us do a bunch of examples to set these ideas into place.

Let us do a 2 by 2 example first.

So, consider the two vectors 5, 2 and 1, 3 in R2, write the linear combination of these two vectors with unknown coefficients x1 and x2 and equate it to 0, x1 times 5 comma 2

plus x2 times 1 comma 3, 0, 0.

So, now I am using x1 and x2 in place of a1 and a2.

So, we have the system of linear equations 5 x1 plus x2 is 0 and 2 times x1 plus 3 times

x2 is 0.

And we want to know if the only solution for this is x1 is 0, x2 is zero or whether there

are other solutions.

Remember that this being a homogeneous system, the trivial solution always exists.

This is something we have seen even in the previous week.

We have discussed in the previous week.

So, since it is a 2 by 2 matrix, we can look at the corresponding determinant.

So, here the determinant will be enough to check whether or not the only solution is

0, 0.

So, in this case, the determinant is non-zero.

It is an invertible matrix.

So, the determinant, remember, here is 5 times 2, 5 times 3, minus 2 times 1.

So, 15 minus 2 is 13, which is non-zero.

So, since the determinant is non-zero, it is an invertible matrix.

And once you note an invertible matrix, we know that the only solution is the trivial

solution.

So, it has a unique solution x1 is x2 is 0.

So, the upshot is that the vectors 5, 2 and 1, 3 are linearly independent.

So, I hope this is, you have understood how to solve this question of whether two vectors

are linearly independent or not at least in this case.

We are going to do a bunch of examples.

So, here is an example where you have a 3 by 2 matrix.

So, consider the two vectors 1, 2, 0 and 3, 3, 5 in R2.

So, let us also understand what is this three by two?

This means your vectors are in R3 and you have two vectors.

So, your, this is saying you have a 3 by 2 matrix, which means you have two columns.

So, remember, columns correspond to the number of vectors and the size of the vector is the

number of rows.

So, that tells you it is in R3.

So, in general, if you have an m by n matrix, that means you have n columns which is corresponding

to n vectors and they are, the size of the vectors is m, which means they are in Rm.

We studied this a few slides ago.

So, you have these two vectors 1, 2, 0 and 3, 3, 5.

So, before we go ahead, let us note first of all that we already know the linear independence

of two vectors.

Namely, two vectors are linearly independent means that they are not multiples of each

other.

They are two non-zero vectors.

So, here you can see that these are not multiples of each other.

That means they are linearly independent.

So, you already know this fact.

But let us do this from, in terms of our matrix.

So, write the linear combination of these two vectors with unknown coefficients x1 and

x2 and equate it to 0.

So, x1 times 1, 2, 0 plus x2 times 3, 3, 5 is 0, 0, 0.

From here we get a set of system of linear equations by equating the corresponding coordinates.

So, the first coordinate gives you x1 plus 3 times x2 is 0.

The second coordinate gives you 2 times x1 plus 3 times x2 is 0.

And the third coordinate gives you 0 times x1 plus 5 times x2 is 0.

And now, in this case, it is actually easy to check that the only solution is where x1

and x2 are both 0, because the third equation gives you that 5 times x2 is 0 that means

x2 must be 0.

And then you, if you put that into the first equation that tells you x1 must be 0.

And you need not cross check with the second equation, because we know, already know that

x1 and x2 are 0 is a solution, because this is a homogeneous system.

So, the only solution here, we have a unique solution, namely that x1 and x2 are both 0.

So, if you do not find this adhoc principal, this adhoc way of doing it agreeable, then

you can actually do it the standard way.

Namely, you can use Gaussian elimination and we will see soon that there are, we will see

an example soon where we do actually have to use Gaussian elimination.

So, what is the net result?

The net result is what we observed right at the start, because they are not multiples

of each other, namely that the vectors 1, 2, 0 and 3, 3, 5 are linearly independent.

Let us do a 2 by 3 example.

So, what does this mean?

This means I have three vectors in R2.

So, what, let us take the three vectors 1 comma 2, 1 comma 3 and 3 comma 4 in R2.

And remember that for three vectors being linearly dependent is the same as saying that we can write some one of these vectors in as a linear combination of the others.

So, one way of checking this would be to ask if one of these is a linear combination of

the others.

But a more direct method is to take unknown coefficients x1, x2, x3 and then write down

the equation x1 times 1 comma 2 plus x2 times 1 comma 3 plus x3 times 3 comma 4 is 0, 0.

Let us equate the coordinates.

So, if you equate the coordinates, you get 1 times x1 plus 1 times x2 plus 3 times x3

is 0 and you get the 2 times x1 plus 3 times x2 plus 4 times x3 is 0.

Those are your two equations.

This is a system of linear equations.

And now we can use Gaussian elimination.

So, the augmented matrix for this system is the matrix 1, 1, 3.

0 on the right, 2, 3, 4.

0 on the right.

Let us do Gaussian elimination.

So, that means we have to row reduce.

And if you row reduce, it is very easy to check that your solutions are of the form

x1 is minus 5c, x2 is 2 times c and x3 is c, where c is in real numbers.

So, the point is, as c varies, the solutions vary, which means there are infinitely many

solutions.

So, the net result is that the vectors 1 comma 2, 1 comma 3, and 3 comma 4 are linearly dependent.

So, here, let us observe what happened.

You see, you had 3 unknowns and you had 2 equations.

And we have already seen what happens in these cases in the previous week, that if you have

more unknowns than the number of equations, then you always have lots of solutions.

So, keep that in mind and we will make this precise later on.

Finally, let us see a three by three example.

So, that means you have three vectors in R3.

So, let us take the vectors 1, 2, 0, 0, 2, 4 and 3, 0, 0.

So, actually, right away, you can see that these are linearly independent, because if

we can, if they are not that means one of them can be written as a linear combination

of the other two.

And just by observation you can see that is not possible.

But let us go through our usual method.

So, let us take unknown coefficients, x1, x2, x3 and equate that linear combination

to 0.

So, you have x1 times 1, 2, 0 plus x2 times 0, 2, 4 plus x3 times 3, 0, 0, 0, 0, 0.

Let us take the corresponding coordinates on each side.

So, doing that we get x1 plus x2 times 0 plus 3 times x3 is 0, then 2 times x1 plus 2 times

x2 plus 0 is 0, and then 0 times x1 plus 4 times x2 plus 0 times x3 is 0.

So, now, you could either use Gaussian elimination or you can consider, because it is a 3 by

3 case, you can consider the determinant, which is what we do in this solution or you

can just do it by observation, because in the last, if you look at the last equation,

then this is 4 times x2 is 0 that means x2 must be 0.

Then you put that, substitute that in the second equation that will give you x1 is 0.

And then substitute x1 is 0 in the first equation that will give you x3 is 0.

Alternatively, you can look at this matrix.

So, the corresponding matrix is 1, 0, 3, 2, 2, 0, 0, 4, 0.

And note that this has non-zero determinant.

So, from here you can check that, so what is the determinant here, so you have 0 plus

0 plus 4 times 2s, 8 times 3 is 24 minus 0 minus 0 minus 0.

So, the determinant here is 24.

And so this is an invertible matrix that tells us that this system has a unique solution

0, 0, 0.

So, the upshot is that these vectors are linearly independent.

This is, we sort of observed this at the start of the slide and indeed we have explicitly

proved it.

So, now let us address this question about, this comment that I made earlier that remember

we had three vectors in R2 that means we had the 2 by 3 case and suppose now we have more

than two vectors in R2, so suppose we have n vectors in R2, where n is at least 3.

So, what happens?

To check linear independence, we have to check whether the corresponding homogeneous linear

system vx is 0 has a unique solution x is 0.

But on the other hand, since n is bigger than 3, n is at least 3, which is bigger, strictly

bigger than 2 and this is a homogeneous system with more unknowns than equations, so we have

seen in the previous week that Gaussian elimination will yield infinitely many solutions.

So, any set of n vectors in R2, where n is at least 3, so we have three vectors or four

vectors or 20 vectors or 100 vectors that is going to be a linearly dependent set of

vectors.

That is the conclusion.

So, if you have either a single vector or two vectors in R2 only then do they have a

chance of being linearly independent.

More than two vectors, it is always going to be linearly dependent.

So, you can see that these numbers somehow picks up the fact that we are in R2.

So, we are in R2 and this fact is being picked up by linear independence.

This is important and we will see this shortly in our next few videos.

We can generalize this.

So, suppose you are in Rn and you have more than n vectors.

So, for example, if you are in R3 and you have 4 vectors or you are in R4 and you have

5 vectors or let us say you are in R8 and you have 20 vectors, you can make the same

argument as in the previous slide for R2, namely that you will get a system of linear

equations, homogeneous system of linear equations with more unknowns than number of equations,

and hence, it always has a non-trivial solution that will tell you that these are linearly dependent.

So, if you have more vectors, then the n, I mean, what is in Rn, you have more than

n vectors in Rn, then they are always going to be linearly dependent, important point.

So, again, this is picking up that number n in some way.

Let us do an example in R3.

So, consider the four vectors 1, 2, 0, 0, 2, 4, 3, 0, 0 and 1, 2, 3 in R3, so we have

already checked, remember that the first three are linearly independent.

We checked that these three are linearly independent.

So, now we are introducing a new vector 1, 2, 3.

So, let us do our usual thing and write down three equations by equating the corresponding

coordinates.

So, you have equations x1 times 1, 2, 0 plus x2 times 0, 2, 4 plus x3 times 3, 0, 0 plus

x4 times 1, 2, 3, which we equate to 0, 0, 0.

Look at the corresponding coordinates.

So, we get x1 plus 0 times x2 plus 3 times x3 plus 1 times x4 that is 0 and then 2 times

x1 plus 2 times x2 plus 0 times x3 plus 2 times x4 that is 0, and then 0 times x1 plus

4 times x2 plus 0 times x3 plus 3 times x4 that is 0.

So, now, of course, there are lots of 0s here.

And you may be able to manipulate and solve this by hand.

But as we know the general and fastest and cleanest method of doing this is by Gaussian

elimination.

So, we will write on the augmented matrix that gives us 1, 0, 3, 1, 0; 2, 2, 0, 2, 0;

0, 4, 0, 3, 0.

Let us do row reduction.

So, if you do row reduction, we get the augmented matrix 1, 0, 0, one-fourth, 0, 1, 0, three-fourth,

0, 0, 1, one-fourth and the last column is again 0, 0, 0.

So, what have we obtained?

We obtained that the solutions to this are x1 is minus c by 4.

So, remember that we had this idea here of which variables are independent and which variables are dependent.

So, x4 is the independent variable and x1, x2, x3 are dependent.

So, you put x4 as c and once you put x4 as c from the first equation you will get that

x1 is minus c by 4, from the second you will get that x2 is minus 3c by 4, and from the

third you will get that x3 is minus c by 4.

So, as c varies, this is your set of solutions.

So, just as an example you could take c to be, let us say 1 or maybe c to be 4.

So, if you take c to be 4, you get minus 1 times 1, 2, 0 plus minus 3 times 0, 2, 4 plus

minus 1 times 3, 0, 0 plus 4 times 1, 2, 3.

And you can check that this is actually equal to 0.

So, this gives us, so I have given you an explicit example of an equation or a linear

combination of these four vectors, which yields 0, where the coefficients are not all 0.

In fact, in this case, none of them are 0.

But it is enough to have that not all of them are 0.

So, the net upshot is that these 3, these 4 vectors are linearly dependent.

We already knew this.

Because remember that we said, if you have more than three vectors in R3, meaning four

or more vectors in R3, then they are going to be linearly dependent.

And this example shows you why that is happening.

So, finally, let us talk about the relationship with the determinant.

We have seen some examples with the determinant.

So, let us talk about the relationship with the determinant.

So, now, suppose you have a set of n vectors in Rn, which are linearly independent, rather

we want to check whether they are linearly independent.

So, what do we do?

We take those vectors, express them in terms of their coordinates, make the corresponding

matrix V.

So, that is now an n by n matrix, where the jth entry of, sorry, the jth column of that

matrix corresponds to the jth vector vj.

And then you look at the corresponding homogeneous system of linear equations vx is 0 and ask

whether the only solution for this is x is 0.

That will determine whether or not it is linearly independent.

So, if the only solution is 0, then it is linearly independent.

If the only solution is, well, if there are solutions which are non-zero, then it is not

linearly independent, meaning it is linearly dependent.

So, I can check this by looking at whether or not V is invertible.

And to check whether or not V is invertible I can check what is the determinant of V.

So, if the determinant is 0, then these vectors are linearly dependent.

If the determinant is non-zero, then these vectors are linearly independent.

So, this has a unique solution if and only if this vector V is, this matrix V is invertible,

so this is not A but this is V, should have been V and if A is invertible let us recall

that there exists A inverse such that A times A inverse is 1, meaning the identity, it should be identity is A inverse times A and so the determinant is non-zero.

And we can reverse this, remember, by, if you recall how we went the other way, if the

determinant is non-zero then you can look at the miners and look at the adjugate matrix.

And if the determinant, so that is how you go the other way, this is V and this is V.

So, let us do an example.

So, let us look at these three vectors in R3 1, 4, 2, 0, 4, 3 and 1, 1, 0.

So, the corresponding matrix is 1, 4, 2, 0, 4, 3 and 1, 1, 0.

Well, I should not say that way.

I have always been saying it along the rows.

So, the corresponding matrix is 1, 0, 1, 4, 4, 1 and 2, 3, 0.

So, what we do is, we put this 1, 4, 2 into the first column and 0, 4, 3 into the second

column and 1, 1, 0 into the third column.

And then, let us look at the determinant of V. Apologies, this should be V.

So, this determinant is, let us see what it is, so it us 0 plus 0 plus 4 times 3 times

1 is 12 and then minus 8 minus 3 minus 0.

So, 12 minus 8 minus 3 that is 1.

So, the determinant is 1.

So, this matrix V is invertible.

And so the vectors 1, 4, 2, 0, 4, 3 and 1, 1, 0 are linearly independent.

So, this is an example of the previous idea where if you have n vectors in Rn, you can

deduce whether or not they are linearly independent just by putting them into a matrix, by putting

the jth vector into the jth column and creating that matrix and then looking at the determinant.

If the determinant is 0, then they are linearly independent, if the determinant is 0, they

are linearly dependent.

And if the determinant is non-zero, they are linearly independent.

So, let us summarize what we have seen in this video so far.

So, we have seen in this video linear independence deduces to checking a system of linear equations

where you take the coefficients to be the unknowns and you take the matrix to be, the matrix coming from the vectors by putting the jth vector into the jth column.

So, all this is, of course, for Rm.

If you are in some other vector space, then what I am saying does not work, then you have

to just do that by hand.

So, we will see examples of that later on or in the, and in the tutorials.

So, and then you look at the system of linear equations vx equals 0.

And if that system has a non-trivial solution is a homogeneous system.

In fact, it has a non trivial solution, meaning if there is a solution where x is not 0, then

the set of vectors is linearly dependent.

If the only possible solution is a trivial solution, namely x is 0, then the set of vectors

v1, v2, vn is linearly independent.

And note that if the number of vectors is bigger than the vector space in which one

is working in, meaning the exponent of the vector space, meaning if you have Rn and if you have more than n vectors, so you have n plus 1 or n plus 2 and so on, if you have

that many vectors, so bigger than or equal to n plus 1 vectors, then they are always

linearly dependent.

And if you have exactly n vectors, so you have n vectors in Rn, then you can check the

linear dependence or independence by looking at the determinant of that corresponding matrix

V formed by putting the jth vector into the jth column.

So, this is more or less everything that we have discussed in this video.

Thank you.